Preliminary

Let V be a vector space.

Def. Suppose $V = W_1 \oplus W_2$, and define a map

$$T : V \to V ; x_1 + x_2 \mapsto x_1 \quad (x_1 \in W_1, x_2 \in W_2)$$

It is called projection along W_2 onto W_1.

Lemma. Let $T : V \to V$ be a projection along W_2 onto W_1.

1). T is linear operator
2). $W_1 = \operatorname{Im} T = \ker(T - I)$
3) $W_2 = \ker T$.
4). $T^2 = T$.

Proof. Let $x, y \in V$ and $\alpha \in F$ be given. Since $V = W_1 \oplus W_2$,

$$\exists!\, x_1, y_1 \in W_1,\ x_2, y_2 \in W_2 \ \text{s.t.}\ x = x_1 + x_2 \text{ and } y = y_1 + y_2.$$

proof of 1).

$$T(\alpha x + y) = T(\alpha x_1 + \alpha x_2 + y_1 + y_2) = T(\alpha x_1 + y_1 + \alpha x_2 + y_2) = \alpha x_1 + y_1 = \alpha T(x) + T(y).$$

proof of 2),

$$\begin{aligned} \operatorname{Im} T &= \{ T(x_1 + x_2) \mid x_1 \in W_1, x_2 \in W_2 \} \\ &= \{ T(x_1) \mid x_1 \in W_1 \} \\ &= \{ T(x) \mid x = T(x) \} \\ &= \{ x \mid T(x) = x \} \end{aligned}$$

Proof of 3). Clearly $W_2 \subseteq \ker T$. To show $W_2 \supseteq \ker T$, let $x \in \ker T$ be given.
If $x \notin W_2$, then $x = x_1 + x_2$ for some $x_1 \in W_1$, $x_2 \in W_2$ s.t. $x_1 \neq 0$.
But, then $T(x) = T(x_1 + x_2) = x_1 = 0$. Contradiction.

proof of 4). Let $x = x_1 + x_2 \in V$ where $x_1 \in W_1$, $x_2 \in W_2$ be given.

$$T^2(x) = T(x_1) = x_1 = T(x).$$

「In general, for linear map $T: V \to V$, $\ker T \cap \operatorname{Im} T \neq \{0\}$.
For instance, let

$$T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$ be a Matrix.

Then,

$$\ker T = \left\{ a \begin{pmatrix} 1 \\ 0 \end{pmatrix} \middle| a \in F \right\} \text{ and } \operatorname{Im} T = \left\{ b \begin{pmatrix} 1 \\ 0 \end{pmatrix} \middle| b \in F \right\},$$

That is, these are same. $\ker T = \operatorname{Im} T$.

「Counterexample for $W_1 \oplus W_2 = W_1 \oplus W_2'$ but $W_2 \neq W_2'$.
Let $V = F^2$, and $W_1 = \langle (1,0) \rangle_F$.
Put $W_2 = \langle (1,1) \rangle_F$ and $W_2' = \langle (1,-1) \rangle_F$.

「Lemma. $T: V \to V$ be a linear on fin. dim V, then

$V = \ker T + \operatorname{Im} T$ or $\ker T \cap \operatorname{Im} T = \{0\}$ iff $V = \ker T \oplus \operatorname{Im} T$.

Lemma. Let V be a vector space, $T: V \to V$ be a linear operator.

$T^2 = T$ if and only if T is a projection along $\ker T$ onto $\ker(T-I)$.

Proof. Suppose $T^2 = T$. Let $x \in V$ be given. Then,

$$x = x - T(x) + T(x)$$

where $x - T(x) \in \ker T$ and $T(x) \in \ker(T-I)$ because

$T(x - T(x)) = T(x) - T^2(x) = 0$ and $(T-I)(T(x)) = T^2(x) - T(x) = 0$.

Moreover, let $y \in \ker T \cap \ker(T-I)$. Then

$$0 = T(y) = y$$

Therefore $\ker T \cap \ker(T-I) = \{0\}$, so

$$V = \ker T \oplus \ker(T-I)$$

Let $x_1 \in \ker T$ and $x_2 \in \ker(T-I)$ be given. Then

$$T(x_1 + x_2) = T(x_2) = x_2.$$

Consequently, T is a projection. The converse holds for general projections.

Let $V = W_1 \oplus W_2$, and $T: V \to V$ be a projection along W_2 onto W_1. For $x_1 \in W_1$, $x_2 \in W_2$

$$T(x_1 + x_2) = x_1 = T(x_1) = T^2(x_1) = T^2(x_1 + x_2)$$

Thus $T = T^2$.

⌈Def. Let $(V, \langle\cdot\rangle)$ be an inn. prod. space.

A Projection $T: V \to V$ is called orthogonal projection if;

$$(\ker T)^{\perp} = \operatorname{Im} T \quad \text{and} \quad (\operatorname{Im} T)^{\perp} = \ker T$$

⌈Thm. Let $T: V \to V$ be a linear operator on inner product space $(V, \langle\cdot\rangle)$.

T is orthogonal projection if and only if adjoint T^* exists, and $T^2 = T = T^*$

Proof. Since T is projection iff $T^2 = T$, enough to show that

$$(\ker T)^{\perp} = \operatorname{Im} T \text{ and } (\operatorname{Im} T)^{\perp} = \ker T \quad \text{iff} \quad T^* \text{ exists and } T = T^*.$$

Let $x, y \in V$ be given. Since $V = \ker T \oplus \operatorname{Im} T$, for some $x_1, y_1 \in \ker T$ and $x_2, y_2 \in \operatorname{Im} T$,

$$x = x_1 + x_2 \quad \text{and} \quad y = y_1 + y_2.$$

Now,

$$\langle x, T(y)\rangle = \langle x_1 + x_2, T(y_1 + y_2)\rangle = \langle x_1 + x_2, y_2\rangle = \langle x_1, y_2\rangle + \langle x_2, y_2\rangle = \langle x_2, y_2\rangle$$

$$\langle T(x), y\rangle = \langle T(x_1 + x_2), y_1 + y_2\rangle = \langle x_2, y_1 + y_2\rangle = \langle x_2, y_1\rangle + \langle x_2, y_2\rangle$$

These are same. Therefore, $T = T^*$.

Conversely, let $x \in \ker T$ and $y \in \operatorname{Im} T$ be given. Then,

$$\langle x, y\rangle = \langle x, T(y)\rangle = \langle T^*(x), y\rangle = \langle T(x), y\rangle = \langle 0, y\rangle = 0. \qquad W \subseteq W^{\perp\perp}$$

Therefore $\ker T \subseteq (\operatorname{Im} T)^{\perp}$ and $\operatorname{Im} T \subseteq (\ker T)^{\perp}$

To show $(\ker T)^{\perp} \subseteq \operatorname{Im} T$, let $y \in (\ker T)^{\perp}$. Then

$$\|y - T(y)\|^2 = \langle y, \underbrace{y - T(y)}_{\in \ker T}\rangle - \langle T(y), y - T(y)\rangle$$

$$= -\langle y, T^*(y - T(y))\rangle$$

$$= -\langle y, T(y) - T^2(y)\rangle = 0$$

Therefore $y = T(y) \in \ker(T - I) = \operatorname{Im} T$, $(\ker T)^{\perp} = \operatorname{Im} T$.

Now, $\ker T \subseteq (\ker T)^{\perp\perp} = (\operatorname{Im} T)^{\perp}$

「Thm. Let V be a fin.dim inn.prod space over F, and let $T : V \to V$ be a linear map.
Assume T has distinct eigenvalues $\lambda_1, \cdots, \lambda_k \in F$.
Assume T is Normal if $F = \mathbb{C}$, T is Herimitian if $F = \mathbb{R}$.
For each $1 \le i \le k$, define

$$T_i : V \to V : \sum_{j=1}^{n} x_j \mapsto x_i \quad \text{where } x_j \in E_{\lambda_j}$$

Then, the followings hold
1) $T_i \circ T_j = \delta_{ij} T_i$, for any $1 \le i, j \le k$
2) $I = T_1 + \cdots + T_k$
3) $T = \lambda_1 T_1 + \cdots + \lambda_k T_k$.

「Corollary.
If $F = \mathbb{C}$,

T is normal if and only if $g(T) = T^*$ for some $g(t) \in \mathbb{C}[x]$.

proof.

$$g(t) = \sum_{i=1}^{k} \overline{\lambda_i} f_i(t) \quad \text{where } f_i(t) = \prod_{j \ne i}^{k} \frac{(t - \lambda_i)}{(\lambda_j - \lambda_i)}$$

Then, $g(\lambda_i) = \overline{\lambda_i}$ for all $1 \le i \le k$. Denote $g(t) = \sum_{i=0}^{k-1} a_i t^i$.

$$\begin{aligned}
g(T) &= a_{k-1} T^{k-1} + a_{k-2} T^{k-2} + \cdots + a_1 T + a_0 \\
&= a_{k-1} (\lambda_1^{k-1} T_1 + \cdots + \lambda_k^{k-1} T_k) \\
&\quad + a_{k-2} (\lambda_1^{k-2} T_1 + \cdots + \lambda_k^{k-2} T_k) \\
&\quad \vdots \\
&\quad + a_1 (\lambda_1 T + \cdots + \lambda_k T_k) \\
&\quad + a_0 \\
&= g(\lambda_1) T_1 + \cdots + g(\lambda_k) T_k \\
&= \overline{\lambda_1} T_1 + \cdots + \overline{\lambda_k} T_k \\
&= \overline{\lambda_1} T_1^* + \cdots + \overline{\lambda_k} T_k^* \\
&= T^*.
\end{aligned}$$

The converse is trivial.

Corollary. In $F=\mathbb{C}$,

T is unitary if and only if T is normal and $\mathrm{ch}(T)(\lambda)=0 \Rightarrow \lambda=\pm 1$.

T is Hermitian if and only if T is normal and all eigenvalues are real.